

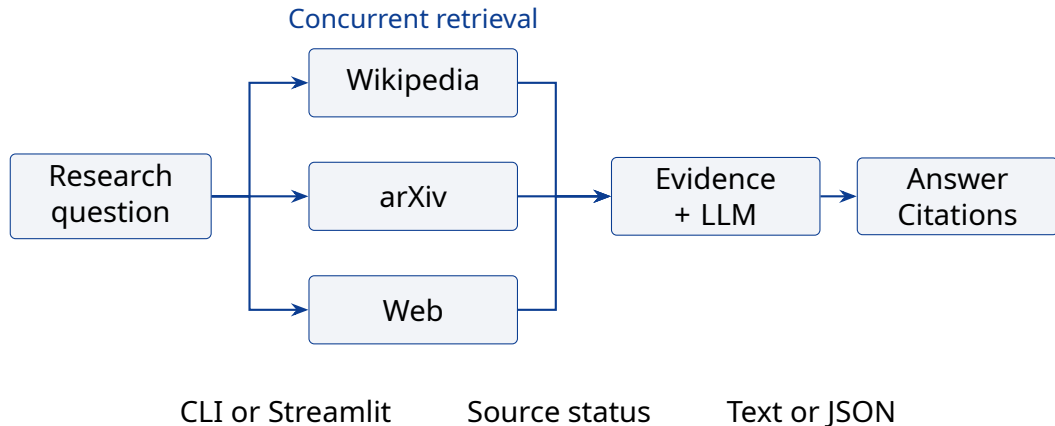
AI Academy, National AI Center

Research Assistant

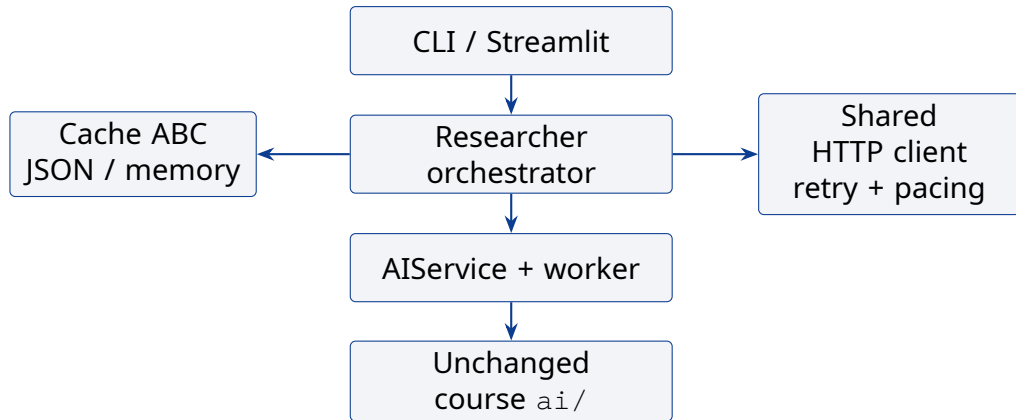
VerteX

Yaqub Xəlilli Səbuhi Xamiyev Aqil Əsgərov

One question, multiple sources

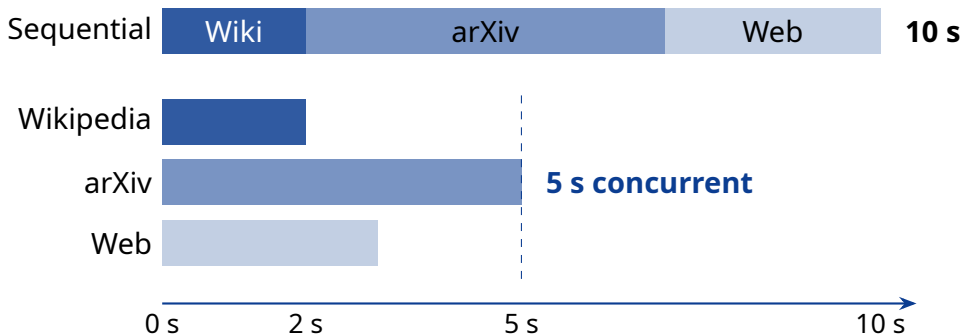


Architecture



Pydantic models validate data across boundaries.

Why async retrieval?



Illustrative timings. Semaphore limits active work. LLM synthesis follows retrieval.

Measured retrieval speedup

Sequential



Concurrent



3.01×

retrieval speedup

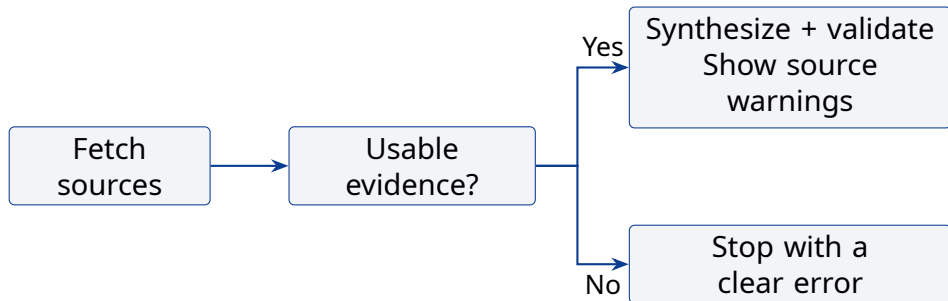
5 questions

3 repetitions

0 failed / empty

Simulated HTTP: 150 ms per source. Median times, cache bypassed.
Retrieval only. These results do not measure live LLM latency.

Failure handling



Slow source

Deadline bounds the wait

Uncited answer

Regenerate within the retry budget

Demonstration modes

Offline sample

5 prepared topics

No network or API key

Explicitly simulated evidence

Live research

External source requests

Configured LLM key

Latency and quota vary

```
python -m researcher demo --offline --no-cache
```

Live demo: inspect references and source status in Streamlit.

Verification evidence

121

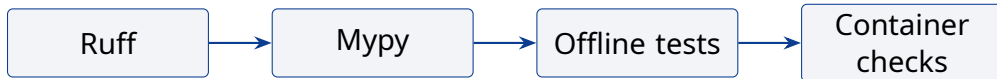
tests passed

93.85%

application coverage

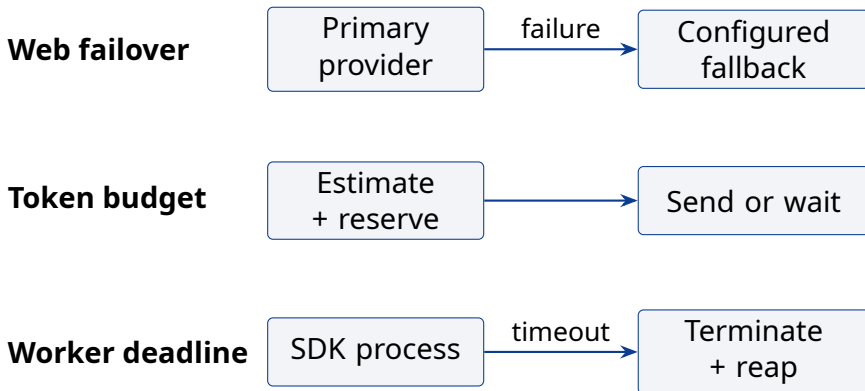
16

course smoke tests



Recorded full run: 17 September 2026. Smoke tests are included in 121. Tests mock external services. Coverage measures execution, not correctness.

Reliability extensions



Failover needs configured providers. Token estimates apply to one service instance.

Live demonstration

Async Research Assistant

Scan to open the deployed app.

VerteX

Questions and discussion



vertex-research-assistant.onrender.com

github.com/yagubbx/Research-Assistant